THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

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Seat Number __________
Student Number |__|__|__|__|__|__|__|__|
Family Name ____________________
First Name ____________________

School of Mathematics & Physics
Semester One Examinations, 2023
MATH1061 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•This is a Closed Book examination - no written materials permitted
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions
•This examination paper will be released to the Library

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

None

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Complete all examination questions in the space provided on this examination paper. If more space is required, use the backs of pages. The final page is a formula sheet, which you may detach. Nothing written on the formula sheet will be marked. Answers will be marked based on correctness, quality of expression, depth of understanding demonstrated, and clarity of explanation. If you have any queries about a particular examination question, or if you make any assumptions about the meaning of an examination question, please state this clearly at the start of your solution to the examination question.

For Examiner Use Only

Question	Mark
1	
2	
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Total _________

Out of 70

Question 1. Let p and q be statement variables.

(a) Use a truth table to show that $\sim p \vee (p \wedge q) \equiv p \to q$. **(2 marks)**

(b) Define a new logical connective $\Box$ as $p \,\Box\, q \equiv p \to \sim q$. Use the Laws of Logical Equivalence, the equivalence of $\to$ to a disjunction, and the definition of $\Box$ to show that

$$(p \,\Box\, q) \,\Box\, p \equiv p \to q.$$

Show your working and name the laws that you use at each step.

(3 marks)

Question 2. Consider the following statement.

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{Z} \text{ such that } x \leq y.$$

(a) Write the negation of the statement. **(1 mark)**

(b) Determine whether the original statement or its negation is true and briefly explain why. **(2 marks)**

Question 3. Prove the following statement by contradiction. For all real numbers r, s, and n, if r is irrational and s is rational and n is a positive integer, then $n(r - s)$ is irrational.

(5 marks)

Question 4. Consider the sequence $\{b_n\}_{n\geq 1}$ defined recursively as

$$b_1 = 2, \quad b_2 = 6 \quad \text{and} \quad b_k = b_{k-1} + b_{k-2} + \gcd(b_{k-1}, b_{k-2}) \text{ for each integer } k \geq 3.$$

Use (strong) mathematical induction to prove that b_n is even for each integer $n \geq 1$.

(5 marks)

Question 5. Let $A = \{1, 2\}$, $B = \{2, 3\}$ and $C = \{1, 2, 3\}$.

(a) Indicate whether each of the following statements is true or false and provide a brief explanation.

(3 marks)

(i) $A \cap B \in C$.

(ii) $\{A, B\}$ is a partition of C.

(iii) $\mid B - C \mid \leq \mid C - B \mid$.

(b) List the elements of each of the following sets.

(2 marks)

(i) $A \times B$.

(ii) $\mathcal{P}(B)$.

Question 6. Let $\mathbb{Z}^+$ be the set of positive integers and $\mathbb{Q}^+$ be the set of positive rationals. Define a function $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Q}^+$ by

$$f((a, b)) = \frac{a}{b}.$$

(a) Is f injective (one one)? Justify your answer.

(2 marks)

(b) Is f surjective (onto)? Justify your answer.

(2 marks)

(c) Suppose that $g : \mathbb{Z}^+ \times \mathbb{Z}^+ \to \mathbb{Q}^+$ is an onto function. What does this tell us about the cardinalities of $\mathbb{Z}^+ \times \mathbb{Z}^+$ and $\mathbb{Q}^+$?

(1 mark)

Question 7. Recall that $[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$ and $(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$.
Prove that $|[6, 12]| \leq |(6, 12)|$.

(5 marks)

Question 8. Consider the relation R on $\mathbb{Z}^+$ defined by

$$a \, R \, b \leftrightarrow (a+b) \text{ is even.}$$

(a) Prove that R is an equivalence relation.

(4 marks)

(b) How many equivalence classes does R have? Justify your answer.

(2 marks)

Question 9. Consider a relation ρ defined on the set $\{1, 2, 3\}$ such that ρ satisfies each of the following:

- $(2, 3) \in \rho$
- $(3, 2) \in \rho$
- $(1, 1) \notin \rho$
- ρ is transitive
- ρ is not symmetric
- ρ is not antisymmetric.

Draw an arrow diagram for a relation ρ that satisfies these properties.

(2 marks)

Question 10. Define a binary operation $\circ$ on $\mathbb{Z}$ by $x \circ y = 2xy + x + y$. You may assume that multiplication and addition in the integers is associative and commutative.

(a) Is $\circ$ commutative? Justify your answer. **(2 marks)**

(b) Prove that the binary operation $\circ$ has an identity. **(2 marks)**

(c) Is $(\mathbb{Z}, \circ)$ a group? Justify your answer. **(2 marks)**

Question 11. Solve the equation $16x - 13 = 4$ in the field $(\mathbb{Z}_{53}, +, \cdot)$. Hence determine the smallest positive integer y such that $16y - 13 \equiv 4 \pmod{53}$. (*Hint:* $16 \times 10 = 53 \times 3 + 1$.)

(3 marks)

For each part (a), (b) and (c) of Question 12, you do not need to compute an integer answer; you may give a mathematical expression that answers the question.

Question 12. Since mid-2020, new car number plates in Queensland consist of three digits from $0, 1, 2, \ldots, 9$ followed by two letters from the alphabet of 26 letters, followed by a further digit. Number plates may have repeated digits and letters. For example, these are valid number plates under this system:

929 **XY**9 002 **BB**4 339 **ZA**2 124 **IF**0

(a) What is the probability that a randomly chosen number plate made under this system has no repeated digits and no repeated letters?

(3 marks)

(b) Under this system, how many different number plates contain at least two occurrences of the digit 2?

(3 marks)

Question 12 continues over...

(c) Under this system, how many different number plates use only prime numbers as digits or use only vowels (A, E, I, O, U) as letters? (Note that "or" is inclusive or.)

(3 marks)

Question 13. Let $S = \{n \in \mathbb{Z} \mid 1 \leq n \leq 33\}$. Prove that if A is any subset of S containing 23 elements, then there exist three distinct elements $x, y, z \in A$ such that $x \equiv y \equiv z \pmod{11}$.

(3 marks)

Question 14. Let G be the graph with the following adjacency matrix.

$$\begin{array}{c c} & \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \\ e \end{array} & \begin{bmatrix} 0 & 1 & 0 & 0 & 3 \\ 1 & 0 & 2 & 0 & 0 \\ 0 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 3 & 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

(a) Draw the graph G. **(2 marks)**

(b) Does G has an Euler circuit? Does G have an Euler trail? Justify your answers.

(2 marks)

Question 15. Let T be a tree on 12 vertices. Suppose T has exactly three vertices of degree 3 and each of the remaining vertices has degree 1 or 5. Using theorems that were given in lectures, determine the number of vertices of degree 5. Explain your reasoning.

(4 marks)

END OF EXAMINATION

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Logical Equivalences

Given any statement variables p, q and r, a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

Commutative laws	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
Associative laws	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive laws	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity laws	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
Negation laws	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
Double negative laws	$\sim (\sim p) \equiv p$	
Idempotent laws	$p \wedge p \equiv p$	$p \vee p \equiv p$
Universal bound laws	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
De Morgan's laws	$\sim (p \wedge q) \equiv \sim p \vee \sim q$	$\sim (p \vee q) \equiv \sim p \wedge \sim q$
Absorption laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negations of $\mathbf{t}$ *and* $\mathbf{c}$	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Valid Argument Forms

Modus Ponens	$p \to q$ p $\therefore q$	Elimination	(a) $p \vee q$, $\sim q$, $\therefore p$ (b) $p \vee q$, $\sim p$, $\therefore q$
Modus Tollens	$p \to q$ $\sim q$ $\therefore \sim p$	Transitivity	$p \to q$ $q \to r$ $\therefore p \to r$
Generalization	(a) p, $\therefore p \vee q$ (b) q, $\therefore p \vee q$	Proof by division into cases	$p \vee q$ $p \to r$ $q \to r$ $\therefore r$
Specialization	(a) $p \wedge q$, $\therefore p$ (b) $p \wedge q$, $\therefore q$		
Conjunction	p q $\therefore p \wedge q$	Contradiction Rule	$\sim p \to \mathbf{c}$ $\therefore p$

Set Identities For all sets A, B and C that are subsets of a universal set U:

1. Commutative Laws	(a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
2. Associative Laws	(a) $(A \cup B) \cup C = A \cup (B \cup C)$ (b) $(A \cap B) \cap C = A \cap (B \cap C)$
3. Distributive Laws	(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity Laws	(a) $A \cup \emptyset = A$ (b) $A \cap U = A$
5. Complement Laws	(a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$
6. Double Complement Law	$(A^c)^c = A$
7. Idempotent Laws	(a) $A \cup A = A$ (b) $A \cap A = A$
8. Universal Bound Laws	(a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$
9. De Morgan's Laws	(a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
10. Absorption Laws	(a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
11. Complement of U and $\emptyset$	(a) $U^c = \emptyset$ (b) $\emptyset^c = U$
12. Set Difference Law	$A - B = A \cap B^c$

The Quotient-Remainder Theorem Given any integer n and any positive integer d, there exist unique integers q and r such that $n = dq + r$ and $0 \le r < d$.

Unique Factorisation Theorem Given any integer $n > 1$, there exists a positive integer k, distinct prime numbers $p_1, p_2, \ldots, p_k$, and positive integers $e_1, e_2, \ldots, e_k$ such that $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, and any other expression for n as a product of prime numbers is identical to this except perhaps for the order in which the factors are written.

Schröder-Bernstein Theorem For all sets A and B, if $|A| \le |B|$ and $|A| \ge |B|$ then $|A| = |B|$.

Binomial Theorem Given any real numbers a and b, and any nonnegative integer n,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k.$$